

Sri Lanka Institute Of Information Technology



IT3021 – Data Warehousing & Business Intelligence Assignment Report 1

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Introduction

Data warehousing is an essential component of modern business intelligence systems that enables organizations to store and analyze large volumes of historical data for decision-making purposes. A data warehouse integrates data from multiple operational sources, transforms the data into a consistent format, and stores it in a structured schema designed for analytical processing. This allows organizations to identify trends, monitor performance, and generate reports that support strategic planning and operational improvements.

In this project, a data warehouse solution was designed and implemented using **Microsoft SQL Server** as the database platform and **SQL Server Integration Services (SSIS)** as the ETL tool. The selected dataset for this project is the **Daily Food Delivery Orders Dataset**, which contains information about food delivery information, restaurant types, orders, and customers. The dataset represents transactional records generated by a food delivery ordering system making it suitable for building a dimensional data warehouse for analytical purposes.

A **star schema** was designed consisting of a central fact table named **Fact_Orders** and several dimension tables including **Dim_Customer**, **Dim_Payment**, **Dim_Restaurant**, and **Dim_Date**. These dimension tables provide descriptive information that allows users to analyze reservation data from different perspectives. The data warehouse was populated through an ETL process that extracts data from a database and a CSV file, performs necessary transformations such as data conversion and lookups, and loads the data into the staging database and then into the data warehouse.

Additionally, the system includes an **accumulating fact table** feature that tracks the lifecycle of order transactions by recording the transaction creation time, completion time, and processing duration in hours. This enables performance monitoring and supports analytical reporting related to order trends, delivery performance, restaurant demand and customer purchasing behavior. The implemented data warehouse demonstrates key data warehousing concepts including dimensional modeling, ETL processing, and data analysis for business intelligence.

Task 01: Data set selection

Data set

	A	B	C	D	E	F	G	H	I	J
	order_id	order_date	customer_age	restaurant_type	order_value	delivery_distance_km	delivery_time_minutes	payment_method	delivery_partner_rating	order_status
1	1	11/5/2024	62	Indian	497.51	11.07	79	UPI	3.9	Cancelled
2	2	8/20/2024	35	Bakery	232.32	5.83	69	Wallet	2.7	Cancelled
3	3	2/28/2024	34	Italian	540.82	3.61	70	Wallet	3.4	Cancelled
4	4	5/26/2024	65	Cafe	1197.99	3.66	18	Card	4.6	Cancelled
5	5	9/21/2024	40	Indian	947.03	12.08	57	UPI	4.9	Delayed
6	6	3/16/2024	51	Italian	835.75	3.56	85	UPI	3.8	Delayed
7	7	11/20/2024	52	Cafe	771.83	14.37	80	Card	3.7	Delivered
8	8	11/24/2024	52	Chinese	926.2	12.81	19	UPI	4.4	Delayed
9	9	7/23/2024	38	Bakery	548.11	13.54	42	Cash	4.1	Delayed
10	10	7/1/2024	24	Fast Food	177.18	8.15	29	Cash	2.9	Delivered
11	11	9/16/2024	27	Cafe	806.68	5.43	53	Cash	3.1	Cancelled
12	12	8/27/2024	61	Bakery	1193.33	9.68	26	UPI	3.5	Delivered
13	13	1/9/2024	31	Chinese	758.14	1.28	57	Cash	2.9	Delayed
14	14	9/2/2024	47	Cafe	937.07	13.41	88	Wallet	3.8	Cancelled
15	15	8/21/2024	62	Chinese	810.18	12.36	87	Cash	5	Delayed
16	16	7/24/2024	38	Bakery	773.12	9.07	50	Wallet	2.8	Cancelled
17	17	4/9/2024	54	Cafe	816.14	10.95	86	UPI	4.8	Delayed
18	18	4/20/2024	43	Cafe	640.6	4.49	50	Cash	4.1	Delayed
19	19	3/1/2024	54	Indian	193.87	4.4	63	UPI	2.8	Delivered
20	20	8/23/2024	23	Chinese	183.09	14.11	52	Cash	2.9	Cancelled
21	21	10/22/2024	45	Italian	343.21	8.39	77	Cash	4.7	Cancelled
22	22	12/24/2024	57	Cafe	628.52	12.79	50	UPI	4	Cancelled
23	23	7/11/2024	38	Indian	471.61	10.22	33	Cash	4.8	Delivered
24	24	6/12/2024	36	Fast Food	873.9	8.41	20	Wallet	3.8	Cancelled
25	25	3/18/2024	53	Bakery	310.93	10.26	17	Wallet	3.1	Delivered

Dataset: <https://www.kaggle.com/datasets/jayjoshi37/daily-food-delivery-orders-and-delivery-time>

Description

The dataset chosen for this assignment includes 2,600 records and roughly a year's worth of operational transaction data from a food delivery platform. Every record in the dataset corresponds to a customer's order for food delivery.

Data set includes information about

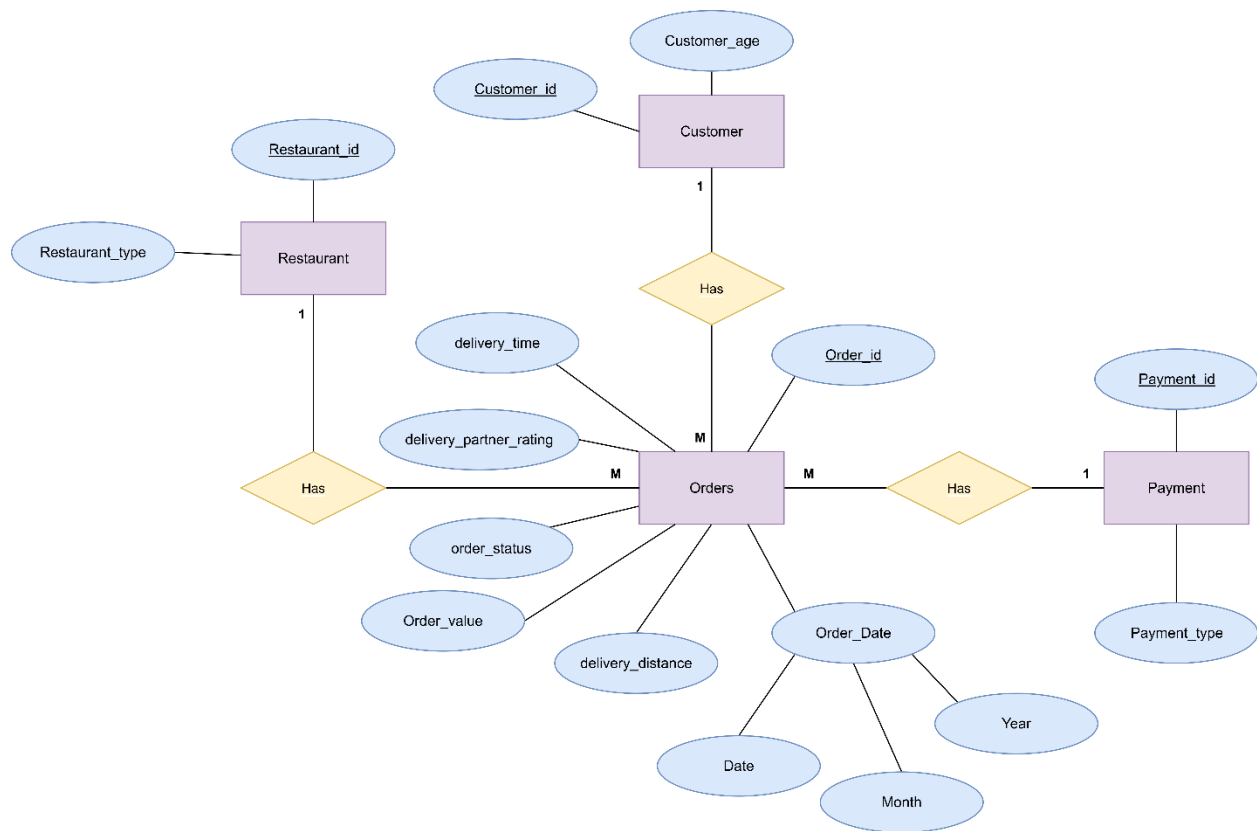
- Order details
- Restaurant category
- Payment method
- Delivery performance
- Customer information

This dataset represents a typical OLTP scenario because each row contains a real world transaction (a food order). The dataset contains sufficient attributes and records to demonstrate data warehouse concepts such as dimensional modeling, ETL processes, and analytical reporting.

Attributes in the data set

- order_id
- order_date
- customer_age
- restaurant_type
- order_value
- delivery_distance_km
- delivery_time_minutes
- payment_method
- delivery_partner_rating
- order_status

ER Diagram



Task 02: Preparation of data sources

The data set was divided into two distinct sources. The ETL pipeline can extract and combine data from various formats, including databases and CSV files, by separating the data into distinct sources.

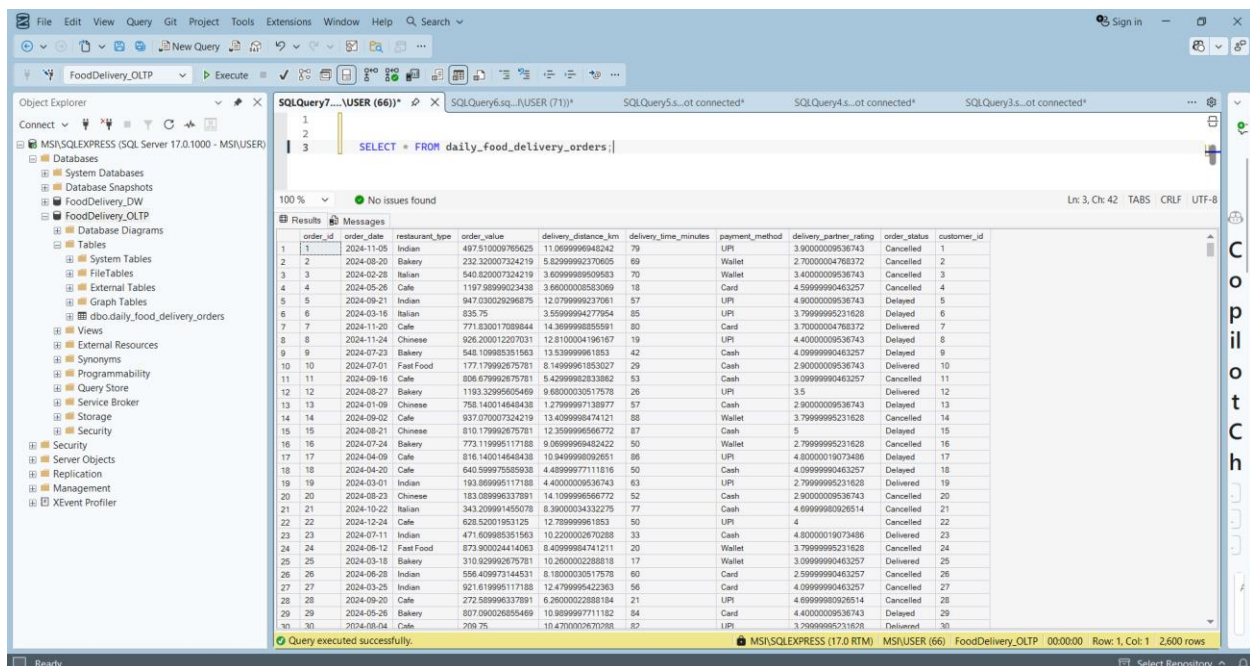
Source 1-SQL Server Database

The primary data source is a SQL Server Database named **FoodDelivery_OLTP**.

The database contains the table **daily_food_delivery_orders**. This table contains operational order transactional data.

Columns included:

- order_id
- order_date
- restaurant_type
- order_value
- delivery_distance_km
- delivery_time_minutes
- payment_method
- delivery_partner_rating
- order_status
- customer_id



The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the 'Object Explorer' with the 'FoodDelivery_OLTP' database selected. The right pane shows a query window with the following SQL query:

```
SELECT * FROM daily_food_delivery_orders;
```

The query results are displayed in a table with the following columns: order_id, order_date, restaurant_type, order_value, delivery_distance_km, delivery_time_minutes, payment_method, delivery_partner_rating, order_status, and customer_id. The table contains 30 rows of data.

order_id	order_date	restaurant_type	order_value	delivery_distance_km	delivery_time_minutes	payment_method	delivery_partner_rating	order_status	customer_id
1	2024-11-05	Indian	487.510009766625	11.0699999845242	79	UPI	3.90000009536743	Cancelled	1
2	2024-08-20	Bakery	232.320007324219	5.82999992370653	69	Wallet	2.70000004768372	Cancelled	2
3	2024-02-28	Italian	540.820007324219	3.60999998905053	70	Wallet	3.40000009536743	Cancelled	3
4	2024-05-26	Cafe	1197.9899923438	3.6800000583069	18	Card	4.59999990463257	Cancelled	4
5	2024-09-21	Indian	947.030026296875	12.0799999237081	57	UPI	4.90000009536743	Delayed	5
6	2024-03-16	Italian	835.75	3.55999994277954	85	UPI	3.79999999231628	Delayed	6
7	2024-11-20	Cafe	771.830017088644	14.3699998855591	80	Card	3.70000004768372	Delivered	7
8	2024-11-24	Chinese	926.200012207031	12.8100004196167	19	UPI	4.40000009536743	Delayed	8
9	2024-07-23	Bakery	548.109985351563	13.539999961853	42	Cash	4.09999990463257	Delayed	9
10	2024-07-01	Fast Food	177.179992675781	8.14999991853027	29	Cash	2.90000009536743	Delivered	10
11	2024-09-16	Cafe	808.679992675781	5.4299999233862	53	Cash	3.09999990463257	Cancelled	11
12	2024-05-27	Bakery	1193.32999505469	9.65000035517578	26	UPI	3.5	Delivered	12
13	2024-01-09	Chinese	758.140014843438	1.27999997138977	57	Cash	2.90000009536743	Delayed	13
14	2024-09-02	Cafe	937.070007324219	13.4099998474121	88	Wallet	3.79999999231628	Cancelled	14
15	2024-08-21	Chinese	810.179992675781	12.3599999656672	87	Cash	5	Delayed	15
16	2024-07-24	Bakery	773.119995117188	9.06999998452422	50	Wallet	2.79999999231628	Cancelled	16
17	2024-04-09	Cafe	816.140014843438	10.9499999292651	86	UPI	4.80000019073486	Delayed	17
18	2024-04-20	Cafe	640.599975855938	4.4899997711816	50	Cash	4.09999990463257	Delivered	18
19	2024-03-01	Indian	193.869995117188	4.40000009536743	63	UPI	2.79999999231628	Delivered	19
20	2024-08-23	Chinese	183.089996337891	14.1099999656672	52	Cash	2.90000009536743	Cancelled	20
21	2024-10-22	Italian	343.209991455078	8.3600003432275	77	Cash	4.69999998026514	Cancelled	21
22	2024-12-24	Cafe	628.52001953125	12.789999981653	50	UPI	4	Cancelled	22
23	2024-07-11	Indian	471.609955351563	10.2200002670288	33	Cash	4.80000019073486	Delivered	23
24	2024-06-12	Fast Food	873.900024414063	8.40999984741211	20	Wallet	3.79999999231628	Cancelled	24
25	2024-03-18	Bakery	310.929992675781	10.2600002288818	17	Wallet	3.09999990463257	Delivered	25
26	2024-06-28	Indian	556.409973144531	8.18000030517578	60	Card	2.59999990463257	Cancelled	26
27	2024-03-25	Indian	921.619995117188	12.4799999422363	56	Card	4.09999990463257	Cancelled	27
28	2024-08-20	Cafe	272.589996337891	6.26000022888184	21	UPI	4.69999998026514	Cancelled	28
29	2024-05-26	Bakery	807.090026855469	10.9899997711812	84	Card	4.40000009536743	Delayed	29
30	2024-08-04	Cafe	208.75	10.47000007670398	82	UPI	3.79999999231628	Delivered	30

Source 2-CSV File

Customer information was stored in a CSV file named customers.csv.

Columns included:

- customer_id
- customer_age

customer_id	customer_age
1	62
2	35
3	34
4	65
5	40
6	51
7	52
8	52
9	38
10	24
11	27
12	61
13	31
14	47
15	62
16	38
17	54
18	43
19	54
20	23
21	45
22	57
23	38
24	36
25	53
26	46

< > customers +

Task 03: Solution architecture

A high-level Data Warehouse and Business Intelligence architecture was designed to integrate data from multiple sources and support analytical reporting. This includes several components that work together to extract, transform, store, and analyze data.

Components of architecture

1.Data Sources

Data sources generate raw data. In this project two data sources are used.

The sources included:

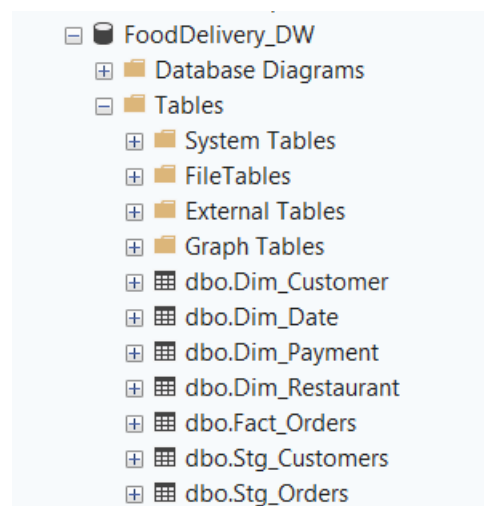
- A SQL Server OLTP database – Contains food delivery order transactions (FoodDelivery_OLTP)
- A CSV file – Contains customer information (customers.csv)

2.ETL Process

The ETL (Extract, Transform, Load) process is implemented using SQL Server Integration Services (SSIS). This process extracts data from the sources, performs, and loads the processed data into the data warehouse

3.Staging Area

The staging area is an intermediate storage layer. This temporarily hold extracted data before it is loaded into the dimensional tables. In this project staging tables (Stg_Customers, Stg_Orders) were created in FoodDelivery_DW database.



4.Data Warehouse

The data warehouse is the central repository that stores integrated and structured data for analytical queries.

- Dimension tables - Dim_Customer, Dim_Restaurant, Dim_Payment, and Dim_Date
- A fact table - Fact_Orders

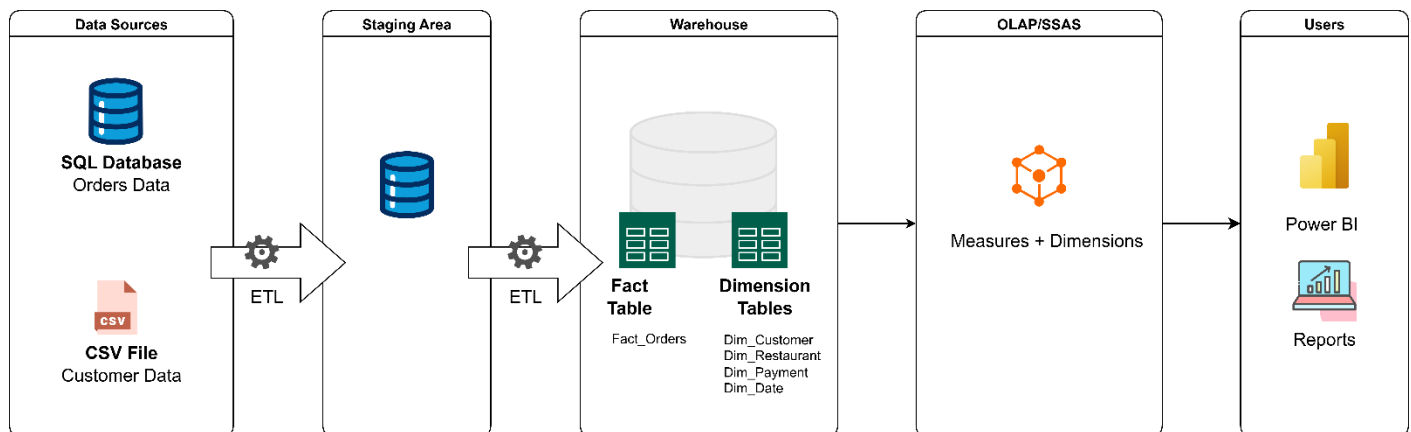
5.OLAP/SSAS Layer

The OLAP layer is used to analyze the data stored in the data warehouse in different ways. Tools like SQL Server Analysis Services (SSAS) can create OLAP cubes, which help users explore the data quickly and easily.

6.Users and Reporting Tools

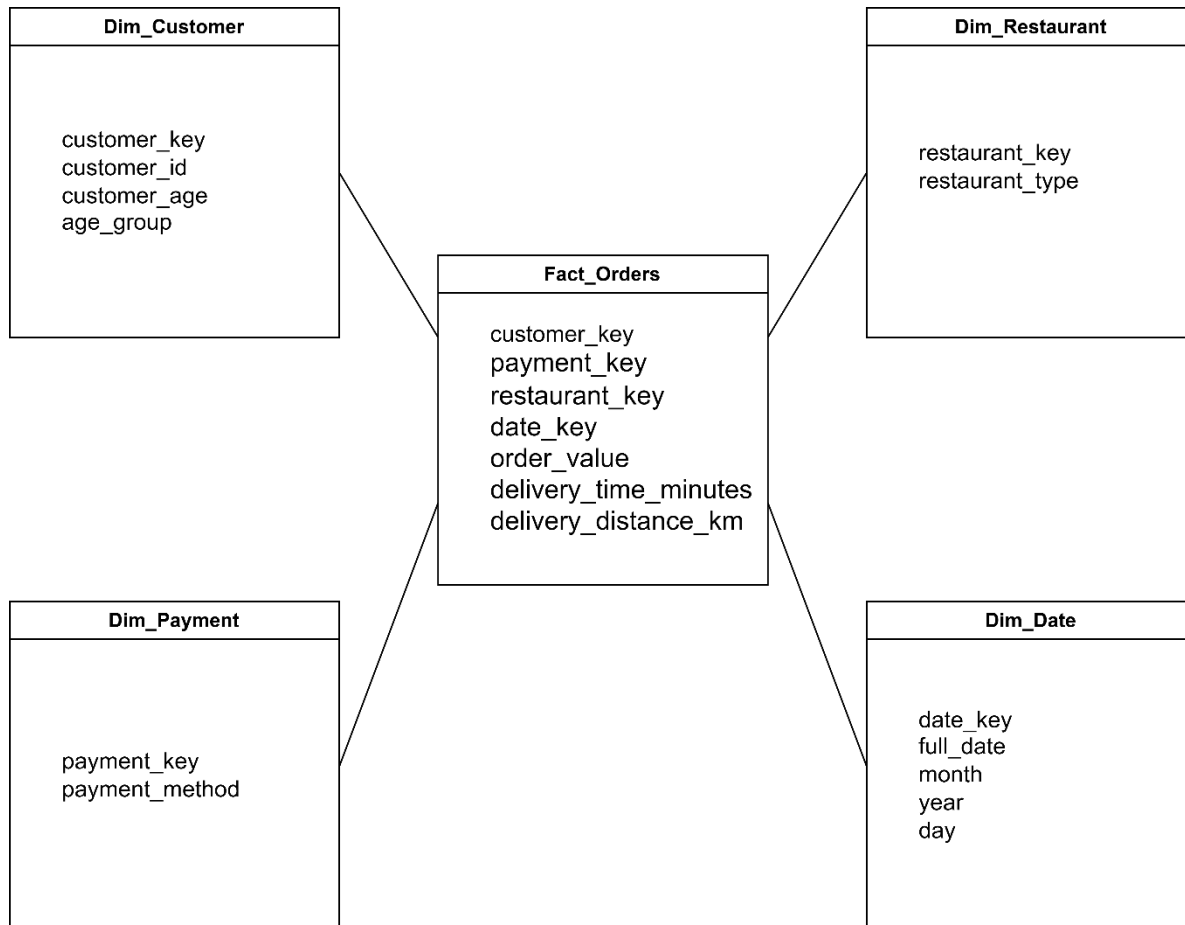
The final layer of the architecture consists of users and reporting tools. Business users and analysts can access the processed data through reporting and visualization tools such as Power BI or Excel. Users can create dashboards, generate reports using these tools.

High level DW & BI solution architecture



Task 04: Data warehouse design & development

Star Schema



Dimensional Model

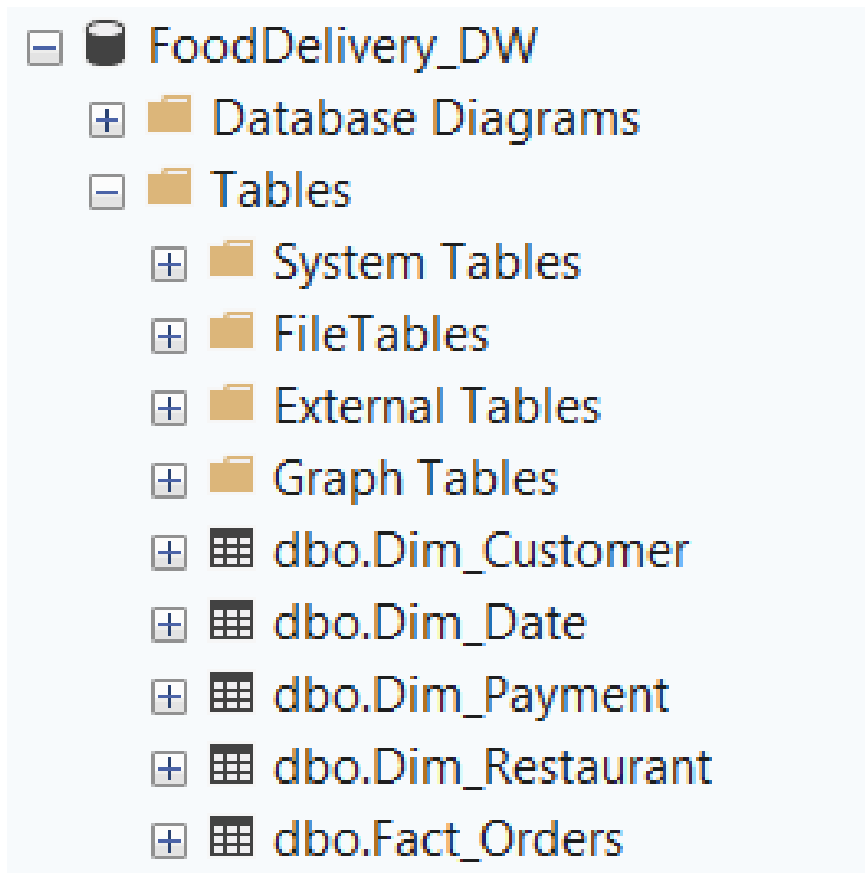
For this project, a **Star Schema** was selected as the dimensional model for the data warehouse. In this design, a central fact table **Fact_Orders** is connected to several dimension tables, `Dim_Customer`, `Dim_Restaurant`, `Dim_Payment`, `Dim_Date` forming a star-like structure that supports efficient data analysis and reporting.

The **Fact_Orders** table stores measurable order transaction data such as order value, delivery distance, delivery time, and delivery partner rating. This table contains foreign keys that link to the dimension tables

Dimension Tables

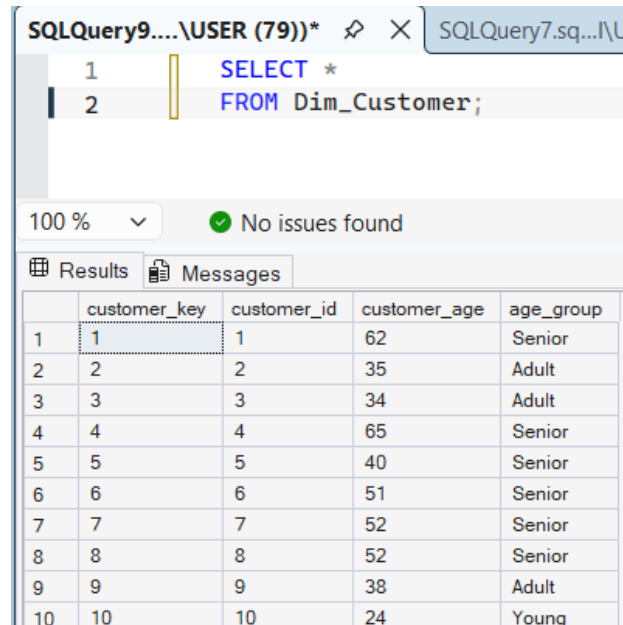
- Dim_Date - This dimension stores time-related attributes derived from the order date, such as day, month, and year
- Dim_Customer - This dimension stores customer-related information including customer ID, customer age, and derived age group
- Dim_Restaurant - This dimension contains restaurant category information such as restaurant type.
- Dim_Payment - This dimension contains payment method information such as credit card, digital wallet, or cash.

SQL Table implementation



Slowly Changing Dimensions

The **Dim_Customer** dimension was implemented as a Slowly Changing Dimension (SCD) to preserve historical changes in customer attributes. When the customer becomes older and the age group changes, a new record can be inserted while maintaining the previous record for historical reference. This approach ensures that historical data remains accurate and supports time-based analysis in the data warehouse.



The screenshot shows a SQL query window titled 'SQLQuery9....\USER (79))*' with the following query:

```
1 SELECT *
2 FROM Dim_Customer;
```

Below the query window, the 'Results' tab is active, displaying a table with 10 rows and 5 columns. The columns are 'customer_key', 'customer_id', 'customer_age', and 'age_group'. The first column is an index from 1 to 10. The data shows a sequence of records for a customer, with the age group changing from 'Senior' to 'Adult' to 'Young' over time.

	customer_key	customer_id	customer_age	age_group
1	1	1	62	Senior
2	2	2	35	Adult
3	3	3	34	Adult
4	4	4	65	Senior
5	5	5	40	Senior
6	6	6	51	Senior
7	7	7	52	Senior
8	8	8	52	Senior
9	9	9	38	Adult
10	10	10	24	Young

(The fact table was also extended in task 06 to include accumulating process tracking columns)

Assumptions Made for the Design

- Each order record represents a single food delivery transaction in the fact table
- Surrogate keys were used for dimension tables
- The Dim_Date dimension was created to enable time-based analysis using attributes such as year, month, and day derived from the order date
- The Dim_Customer dimension was implemented as a Slowly Changing Dimension (SCD)

Task 05: ETL Development

ETL Process

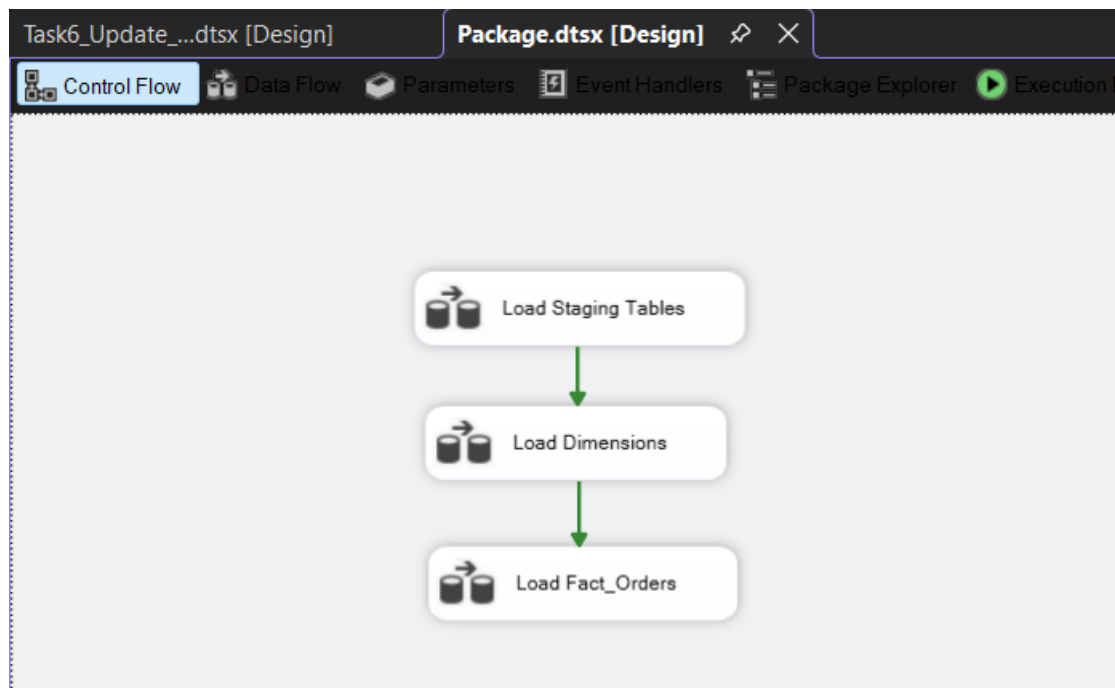
In this project Extract, Transform and Load (ETL) process was implemented using **Microsoft SQL Server Integration Services (SSIS)** to move data from the operational data sources into the data warehouse.

The ETL pipeline performs the following operations:

1. Extract data from multiple heterogeneous sources
2. Load the extracted data into staging tables
3. Transform the data
4. Load the transformed data into dimension tables
5. Load transactional data into the fact table using lookup transformations

Data Sources Used

SQL Server OLTP Database FoodDelivery_OLTP stores transactional operational data and the **customers.csv** file was used as an additional source to demonstrate the ability of SSIS to extract data from external file-based systems.



1:SSIS Control Flow showing the main ETL tasks used to load staging, dimension, and fact tables.

ETL Architecture

Loading the Staging Tables

Extracts raw data from the source systems and loads it into **Stg_Orders**, **Stg_Customers**, **Stg_Restaurants**, **Stg_Payment** staging tables inside the FoodDelivery_DW database. These tables temporarily store the extracted data before it is transformed and loaded into the final dimensional schema

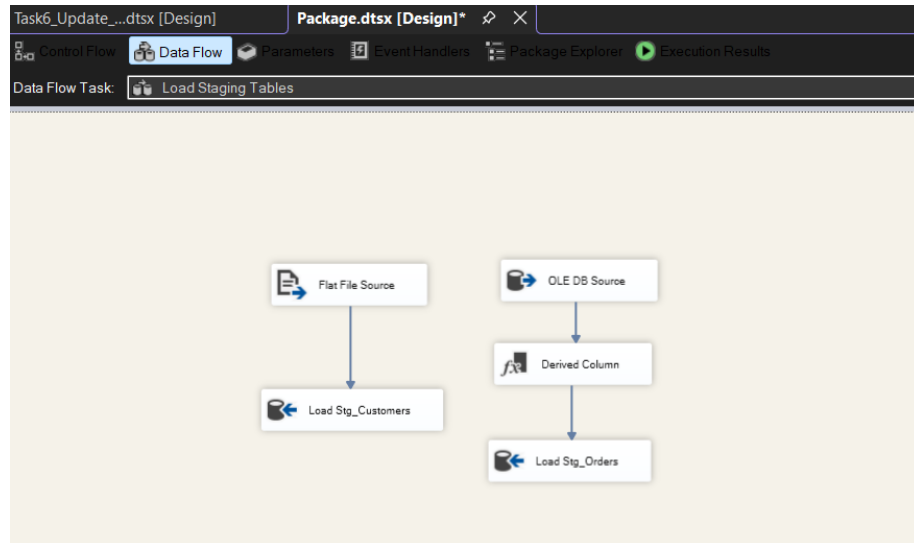


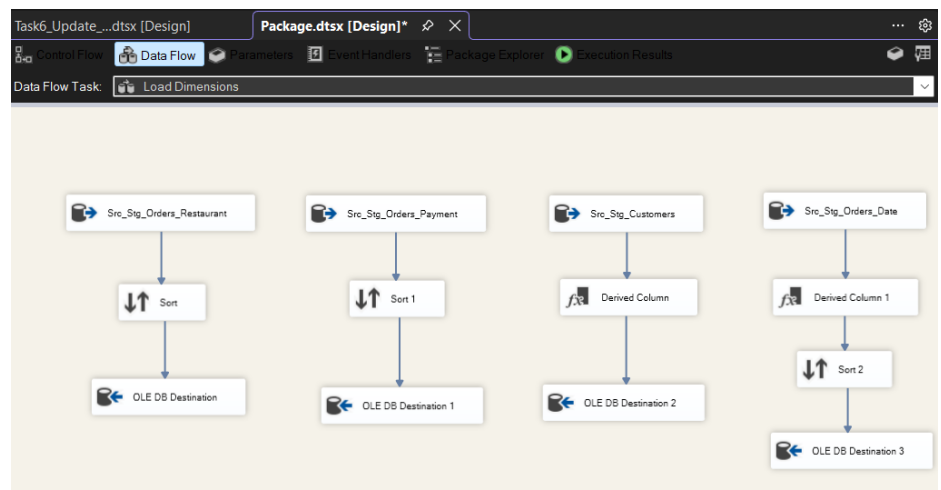
Figure 2: Loading the staging tables

Loading Dimension Tables

Load the **dimension tables** in the data warehouse.

The dimension tables include:

- Dim_Customer
- Dim_Restaurant
- Dim_Payment
- Dim_Date



3: Loading Dimension Tables

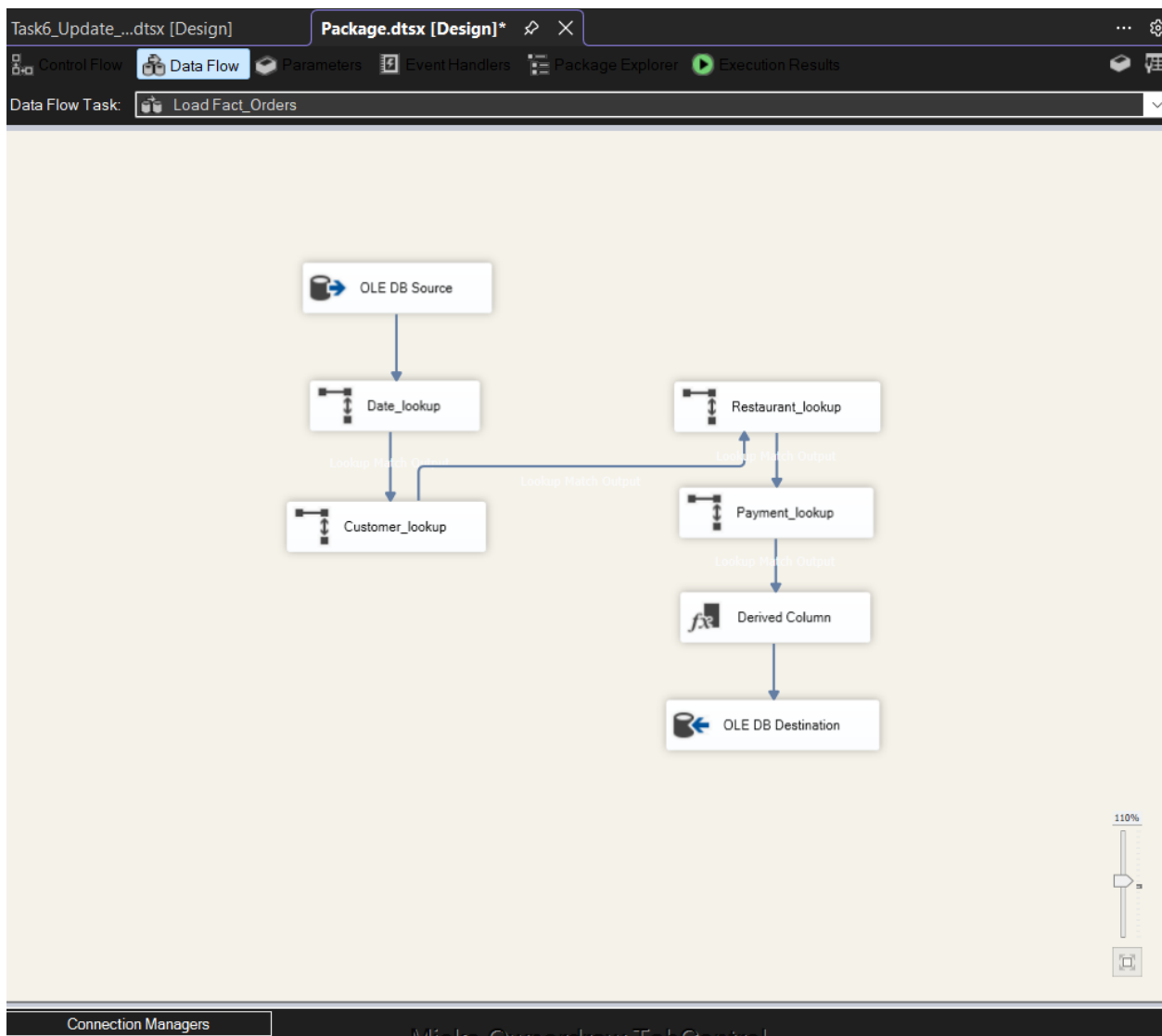
Loading Fact Table

The final stage of the ETL pipeline loads data into the **Fact_Orders** table.

The fact table contains measures related to food delivery transactions such as

- Order_value
- Delivery_time_minutes
- Delivery_distance_km
- Delivery_partner_rating

To maintain referential integrity **Lookup Transformations** were used to retrieve surrogate keys from the dimension tables.

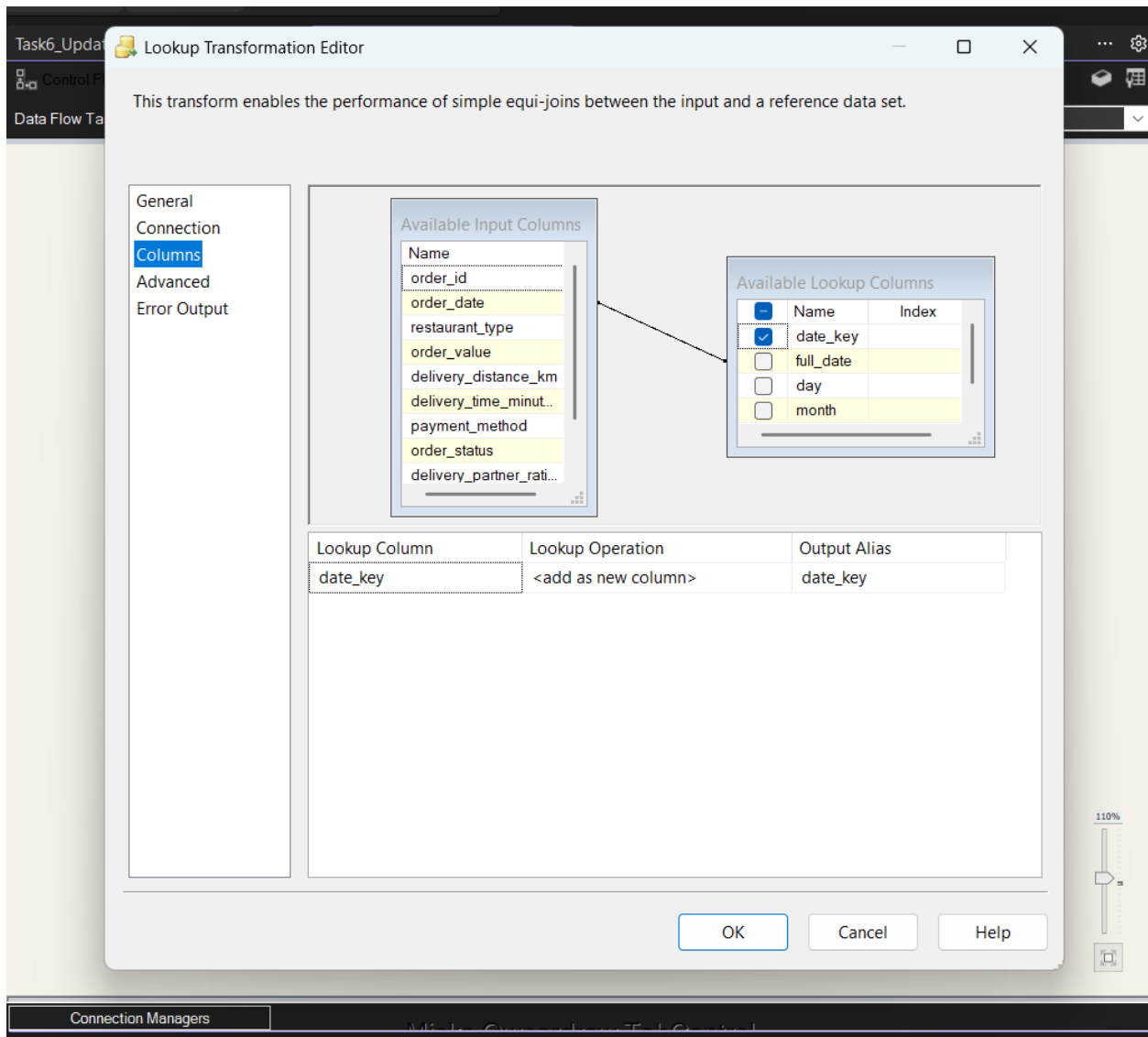


4:SSIS Fact Table Data Flow

Data Transformation

During the ETL process, several transformations were applied.

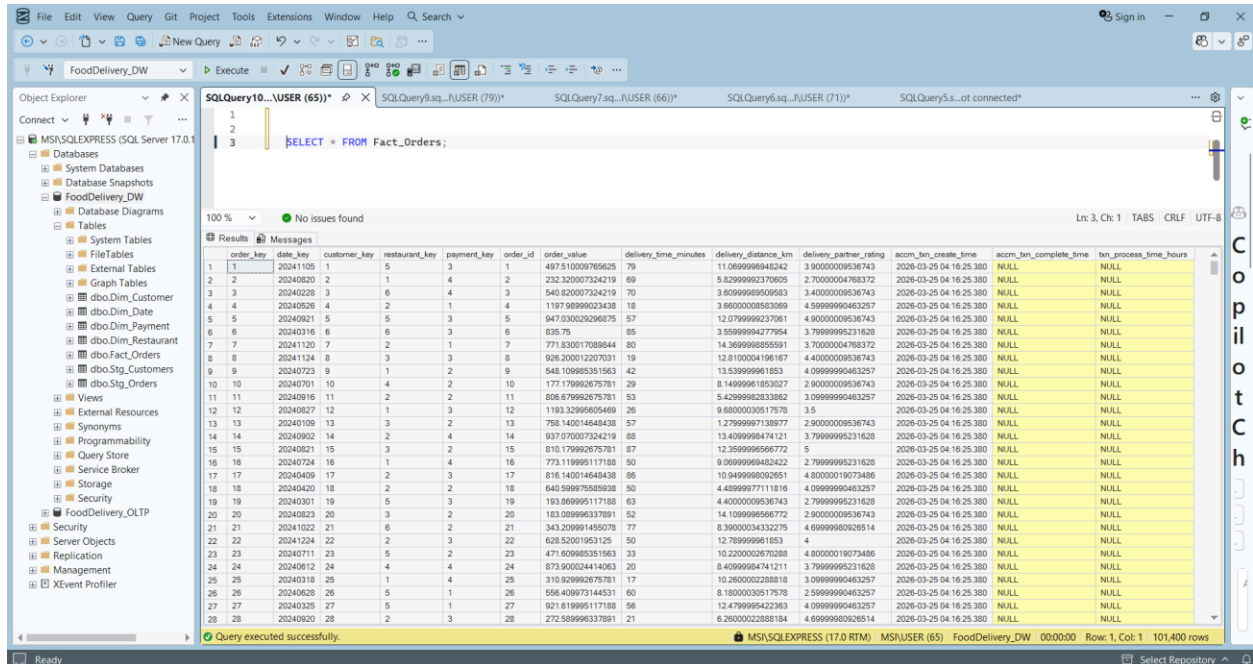
- Mapping staging columns to fact and dimension table columns
- Retrieving surrogate keys using lookup transformations
- Validating matches between staging data and dimension records
- Converting date attributes into dimensional keys



5: Lookup transformation used to retrieve the date surrogate key from the DimDate table

Final Data Load

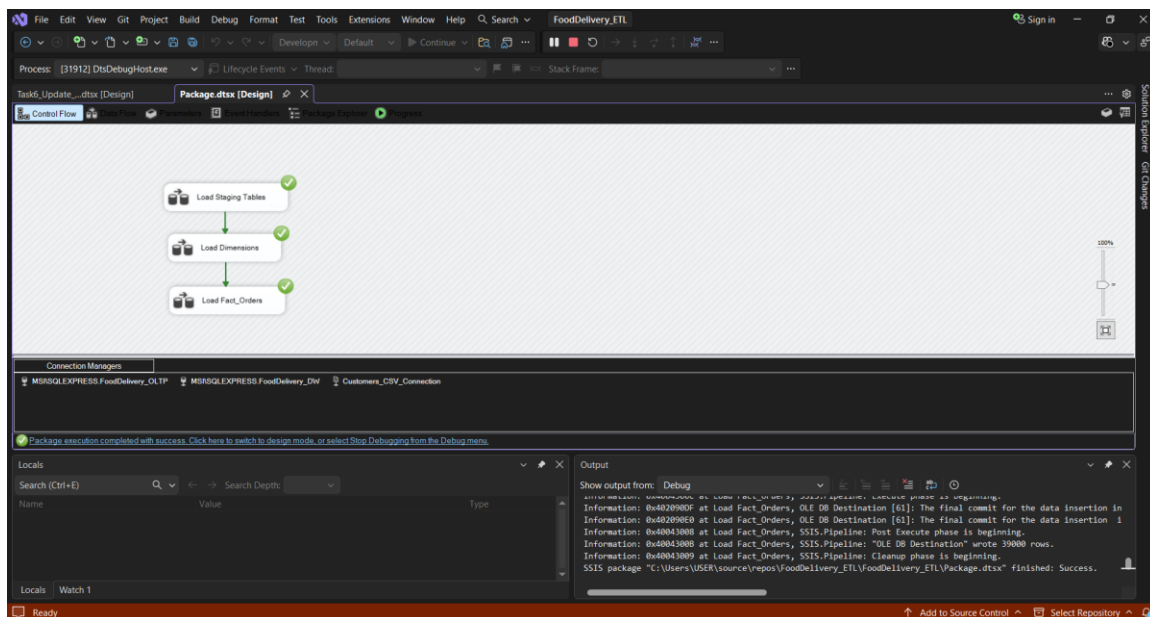
After all transformations and lookups were completed, the final dataset was inserted into the **Fact_Orders** table. The ETL pipeline was successfully executed and validated by querying the data warehouse tables to confirm that the records were loaded correctly.



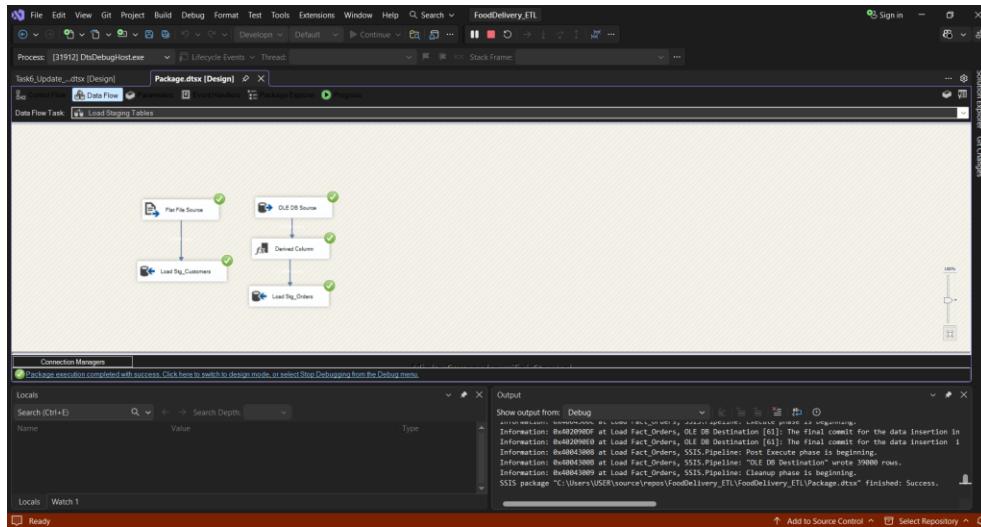
The screenshot displays the SQL Server Enterprise Manager interface. The 'Object Explorer' on the left shows the 'FoodDelivery_DW' database. The 'SQL Query' window in the center shows the executed query: `SELECT * FROM Fact_Orders;`. The 'Results' pane at the bottom shows a table with 101,400 rows and 12 columns. The columns are: `order_id`, `date_key`, `customer_key`, `restaurant_key`, `payment_key`, `order_value`, `delivery_time_minutes`, `delivery_distance_km`, `delivery_partner_rating`, `acqm_ben_create_time`, `acqm_ben_complete_time`, and `ben_process_time_hours`. The data is displayed in a grid format with alternating row colors.

order_id	date_key	customer_key	restaurant_key	payment_key	order_value	delivery_time_minutes	delivery_distance_km	delivery_partner_rating	acqm_ben_create_time	acqm_ben_complete_time	ben_process_time_hours
1	20241105	1	5	3	1	487.51009785625	79	11.0899999945242	3.90000009536743	2026-03-25 04:16:25.380	NULL
2	20240620	2	1	4	2	232.320007324219	69	5.82999992370005	2.70000004763372	2026-03-25 04:16:25.380	NULL
3	20240228	3	6	4	3	540.820007324219	70	3.60999999509583	3.40000009536743	2026-03-25 04:16:25.380	NULL
4	20240526	4	2	1	4	1197.98999023438	18	3.66000000853089	4.59999990463257	2026-03-25 04:16:25.380	NULL
5	20240921	5	5	3	5	947.030029296875	57	12.0799999237061	4.90000009536743	2026-03-25 04:16:25.380	NULL
6	20240316	6	6	3	6	835.75	85	3.55999994277954	3.79999995231628	2026-03-25 04:16:25.380	NULL
7	20241120	7	2	1	7	771.83001708844	80	14.3699999555991	3.70000004763372	2026-03-25 04:16:25.380	NULL
8	20241124	8	3	3	8	926.200012207031	19	12.8100004196187	4.40000009536743	2026-03-25 04:16:25.380	NULL
9	20240723	9	1	2	9	548.109995351963	42	13.539999961853	4.09999990463257	2026-03-25 04:16:25.380	NULL
10	20240701	10	4	2	10	177.179992675781	29	8.14999991853027	2.90000009536743	2026-03-25 04:16:25.380	NULL
11	20240916	11	2	2	11	806.679992675781	53	5.42999992833862	3.09999990463257	2026-03-25 04:16:25.380	NULL
12	20240827	12	1	3	12	1193.32999658489	26	9.800000030517578	3.5	2026-03-25 04:16:25.380	NULL
13	20240109	13	3	2	13	758.140014648438	57	1.27999997138977	2.90000009536743	2026-03-25 04:16:25.380	NULL
14	20240902	14	2	4	14	937.070007324219	88	13.4099999474121	3.79999995231628	2026-03-25 04:16:25.380	NULL
15	20240821	15	3	2	15	810.179992675781	87	12.3599999566772	5	2026-03-25 04:16:25.380	NULL
16	20240724	16	1	4	16	773.119995117188	50	9.06999998482422	2.79999995231628	2026-03-25 04:16:25.380	NULL
17	20240409	17	2	3	17	816.140014648438	86	10.9499999092851	4.80000019073486	2026-03-25 04:16:25.380	NULL
18	20240420	18	2	2	18	640.59975585938	50	4.4099997711816	4.09999990463257	2026-03-25 04:16:25.380	NULL
19	20240301	19	5	3	19	193.869995117188	63	4.40000009536743	2.79999995231628	2026-03-25 04:16:25.380	NULL
20	20240823	20	3	2	20	183.089996337891	52	14.1099999566772	2.90000009536743	2026-03-25 04:16:25.380	NULL
21	20241022	21	6	2	21	343.209991455078	77	8.39000034332275	4.69999990926514	2026-03-25 04:16:25.380	NULL
22	20241224	22	2	3	22	626.52001953125	50	12.789999961853	4	2026-03-25 04:16:25.380	NULL
23	20240711	23	5	2	23	471.609995351963	33	10.2200002670388	4.80000019073486	2026-03-25 04:16:25.380	NULL
24	20240612	24	4	4	24	873.900024414063	20	8.409999904741211	3.79999995231628	2026-03-25 04:16:25.380	NULL
25	20240318	25	1	4	25	310.929992675781	17	10.2600002288818	3.09999990463257	2026-03-25 04:16:25.380	NULL
26	20240628	26	5	1	26	556.409973144531	60	8.18000030517578	2.59999990463257	2026-03-25 04:16:25.380	NULL
27	20240325	27	5	1	27	921.619995117188	56	12.4799999422363	4.09999990463257	2026-03-25 04:16:25.380	NULL
28	20240920	28	2	3	28	272.589996337891	21	6.26000022888184	4.69999990926514	2026-03-25 04:16:25.380	NULL

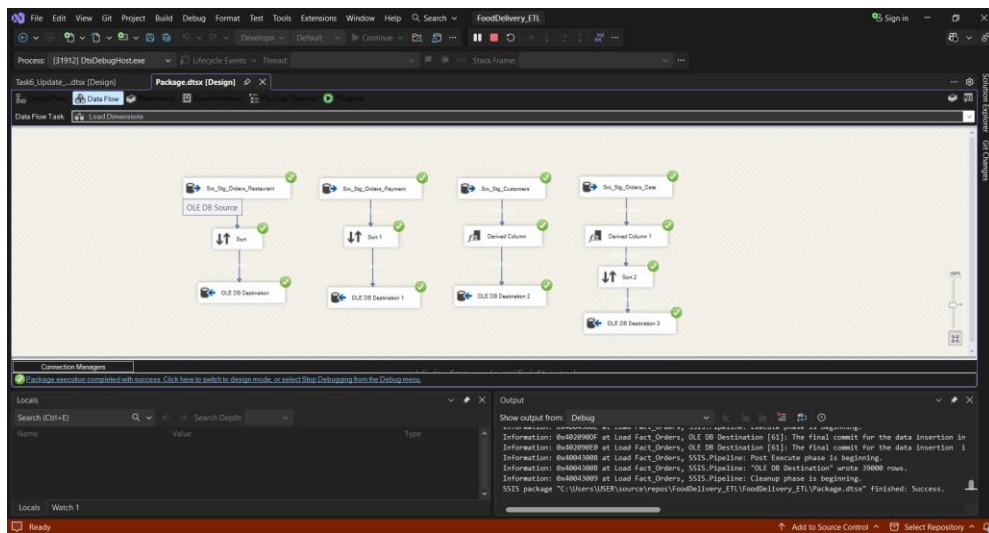
ETL Execution Validation



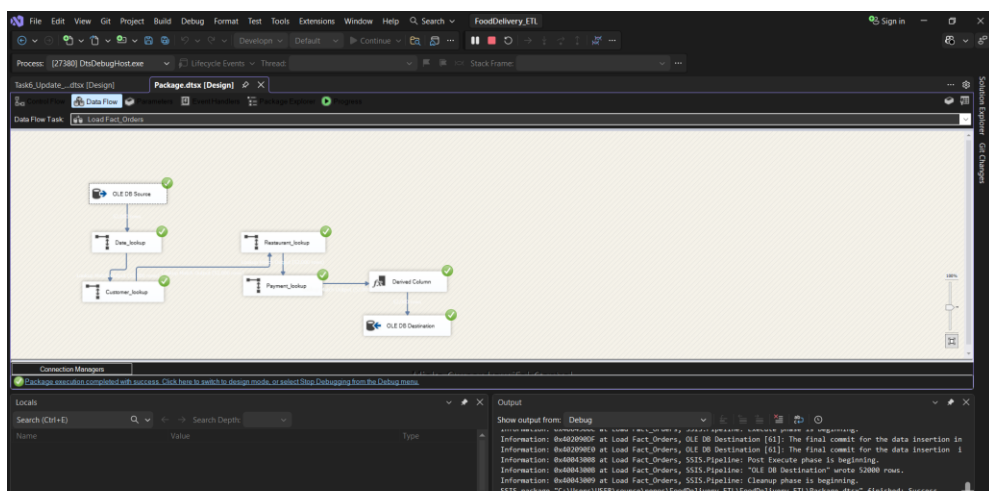
6:control flow with validation



9:Load staging tables data flow validation



8:Load dimintions data flow validation



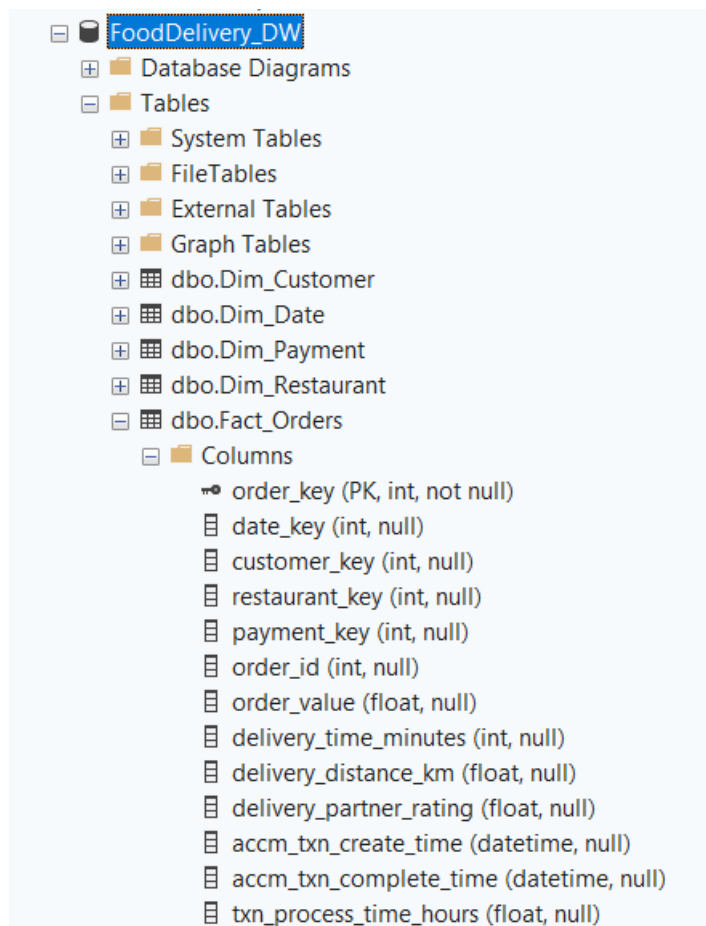
7:Load fact table data flow validation

Task 06: ETL Development - Accumulating Fact Table

Extending the Fact Table

To implement the accumulating fact table concept, additional columns were added to the Fact_Orders table to track the lifecycle of a transaction

- **accm_txn_create_time** – Timestamp when the order record is created warehouse
- **accm_txn_complete_time** – Timestamp when the order transaction is completed
- **txn_process_time_hours** – Total processing duration in hours



Preparing the Completion Dataset

During the ETL process that loads data into the fact table, the column **accm_txn_create_time** is automatically set with the current system date and time. This is the moment the transaction was first recorded in the data warehouse.

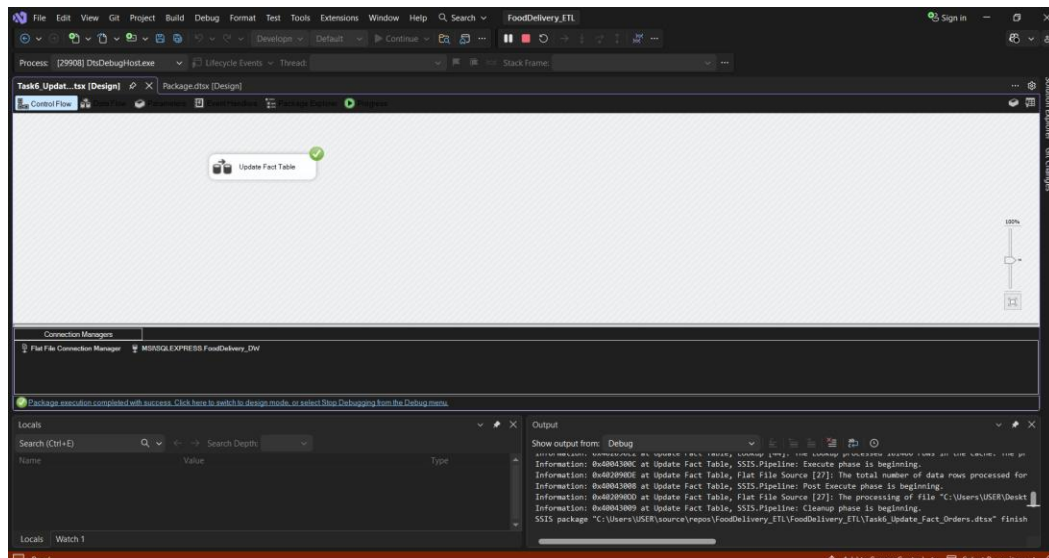
This dataset is used by a separate ETL package named **Task6_Update_Fact_Orders** to update the accumulating fact table.

A	B
txn_id	accm_txn_complete_time
101	3/25/2026 12:30
102	3/25/2026 15:45
103	3/26/2026 9:15

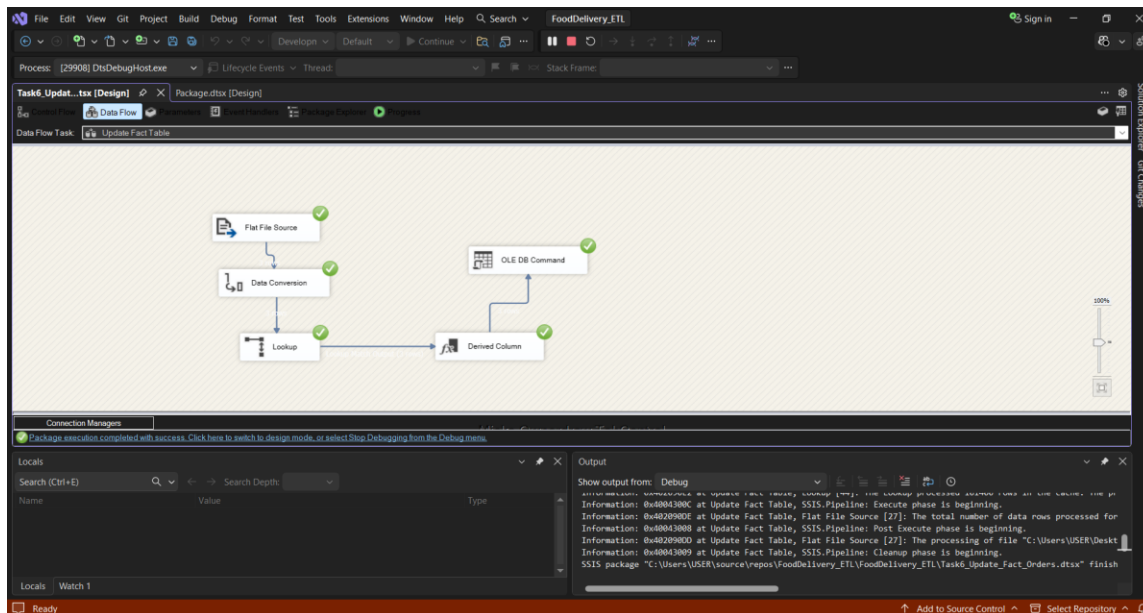
Updating the Accumulating Fact Table

ETL Steps

- Flat File Source reads completion timestamps from the dataset
- Flat File Source reads completion timestamps from the dataset
- Data Conversion ensures correct data type
- OLE DB Command updates the Fact_Orders table



10:successful completion of the accumulating fact update package



11:successful processing of the update pipeline

Validation of Accumulating Fact Updates

After running the update ETL process, validation queries were executed to confirm that the completion time and processing duration were correctly updated

The screenshot shows the SQL query result for 'Fact_Orders'. The query filters for rows where 'accm_txn_complete_time' is not null. The result set shows columns: order_id, accm_txn_create_time, accm_txn_complete_time, and txn_process_time_hours.

order_id	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
1	2026-03-25 04:16:25.380	2026-03-25 12:30:00.000	8
2	2026-03-25 04:16:25.380	2026-03-25 15:45:00.000	11
3	2026-03-25 04:16:25.380	2026-03-26 09:15:00.000	29
4	2026-03-25 04:16:25.380	2026-03-25 12:30:00.000	8
5	2026-03-25 04:16:25.380	2026-03-25 15:45:00.000	11
6	2026-03-25 04:16:25.380	2026-03-26 09:15:00.000	29
7	2026-03-25 04:16:25.380	2026-03-25 12:30:00.000	8
8	2026-03-25 04:16:25.380	2026-03-25 15:45:00.000	11
9	2026-03-25 04:16:25.380	2026-03-26 09:15:00.000	29
10	2026-03-25 04:16:25.380	2026-03-25 12:30:00.000	8
11	2026-03-25 04:16:25.380	2026-03-25 15:45:00.000	11
12	2026-03-25 04:16:25.380	2026-03-26 09:15:00.000	29
13	2026-03-25 04:16:25.380	2026-03-25 12:30:00.000	8
14	2026-03-25 04:16:25.380	2026-03-25 15:45:00.000	11
15	2026-03-25 04:16:25.380	2026-03-26 09:15:00.000	29
16	2026-03-25 04:16:25.380	2026-03-25 12:30:00.000	8
17	2026-03-25 04:16:25.380	2026-03-25 15:45:00.000	11
18	2026-03-25 04:16:25.380	2026-03-26 09:15:00.000	29
19	2026-03-25 04:16:25.380	2026-03-25 12:30:00.000	8
20	2026-03-25 04:16:25.380	2026-03-25 15:45:00.000	11
21	2026-03-25 04:16:25.380	2026-03-26 09:15:00.000	29
22	2026-03-25 04:16:25.380	2026-03-25 12:30:00.000	8

12:SQL query result showing calculated transaction processing hours